

Bike Sharing Demand Analysis

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1. Introduction

1.1 Problem Definition

Bike-sharing systems have become a critical component of sustainable urban transportation networks. With over 500 programs worldwide operating more than 500,000 bicycles, these systems enable users to rent bikes from one location and return them at another. The Capital Bikeshare system in Washington D.C. offers a rich dataset for understanding how weather conditions, temporal patterns, and seasonal effects drive bike rental demand — insights with direct operational value for fleet management and capacity planning.

1.2 Objectives

1. Build accurate predictive models for daily and hourly bike rental demand.
2. Identify the main factors that drive demand using feature importance analysis.
3. Detect non-linear temperature effects and tipping points where demand behavior changes.
4. Compare casual and registered user demand patterns through user segmentation.
5. Apply and benchmark advanced ensemble methods to achieve high prediction accuracy.

1.3 Scope

Dimension	Detail
Dataset	UCI Bike Sharing Dataset (Fanaee-T & Gama, 2014)
Period	January 2011 to December 2012 (731 days, 17,379 hours)
Location	Washington D.C., Capital Bikeshare system
Target Variables	Total rentals (cnt), casual users, registered users
Models Compared	OLS, Ridge, Lasso, PCR, PLS, GAM, Random Forest, GBM, XGBoost, Stacked Ensemble

2. Literature Review

2.1 Background

Fanaee-T and Gama (2014) introduced the UCI Bike Sharing Dataset and demonstrated event detection using ensemble methods, establishing this dataset as a benchmark for demand prediction research.

Yin, Lee, and Wong (2014) applied Random Forest and Support Vector Regression to bicycle sharing demand prediction at Stanford, achieving $R^2 \approx 0.89$ with temporal features — a key benchmark this study aims to surpass.

Gebhart and Noland (2014) studied the Washington D.C. Capital Bikeshare system specifically and found that temperature and precipitation significantly influence trip generation, with optimal conditions around 20–25°C.

Böcker et al. (2013) conducted a systematic review of weather and cycling behavior, identifying temperature thresholds beyond which cycling decreases — suggesting non-linear effects that simple linear models cannot capture.

Ashqar et al. (2017) used gradient boosting to predict bike availability at stations, demonstrating the value of ensemble approaches for urban mobility demand forecasting.

2.2 Gap Identification

Gap	Existing Approaches	Our Contribution
User Segmentation	Model only total count	Separate GAM analysis for casual and registered users
Temperature Effects	Assume linear relationship	Tipping-point detection via GAM first-derivative analysis
Multicollinearity	Drop variables or ignore	PCA on environmental features retaining 99.8% variance
Model Comparison	Single method focus	Comprehensive benchmark of 10+ methods on identical splits
Interpretability	Black-box models only	SHAP analysis with beeswarm and dependence plots

3. Dataset Description

3.1 Data Source

The UCI Bike Sharing Dataset contains two files: day.csv with 731 daily aggregated records and hour.csv with 17,379 hourly records. Both cover the period January 2011 to December 2012 for the Capital Bikeshare system in Washington D.C.

3.2 Variables

Variable	Description	Type
instant	Record index	ID
dteday	Date	Date
season	Season (1=Spring, 2=Summer, 3=Fall, 4=Winter)	Categorical
yr	Year (0=2011, 1=2012)	Binary
mnth	Month (1–12)	Categorical
hr	Hour (0–23)	Categorical (hourly only)
holiday	Holiday indicator	Binary
weekday	Day of week (0–6)	Categorical
workingday	Working day indicator	Binary
weathersit	Weather situation (1=Clear, 2=Mist, 3=Light Precip, 4=Heavy Precip)	Categorical
temp	Normalized temperature (actual/41)	Continuous
atemp	Normalized feeling temperature (actual/50)	Continuous
hum	Normalized humidity (actual/100)	Continuous
windspeed	Normalized wind speed (actual/67)	Continuous
casual	Count of casual users	Target
registered	Count of registered users	Target
cnt	Total count (casual + registered)	Target

4. Data Cleaning and Feature Engineering

4.1 Missing Values and Data Quality

A comprehensive missing-value audit was conducted on both the daily and hourly datasets. No missing values were found across all 17 variables in the daily dataset (731 records) or the hourly dataset (17,379 records). All continuous variables (temp, atemp, hum, windspeed) are normalized to the [0,1] range as documented by the UCI data providers.

4.2 Feature Encoding

Categorical variables were converted to factors with domain-meaningful labels to enable correct model interpretation. Continuous temperature variables were de-normalized to actual Celsius and perceived temperature scales for interpretability:

- season: encoded as Spring, Summer, Fall, Winter
- weathersit: encoded as Clear, Mist, Light Precip, Heavy Precip
- yr: encoded as 2011, 2012
- weekday: encoded as Sun through Sat
- temp_actual = temp × 41 (degrees Celsius); atemp_actual = atemp × 50

The actual temperature range spans approximately 1.6°C to 35.6°C across the two-year period.

4.3 PCA on Environmental Features

The four environmental variables (temp, atemp, hum, windspeed) exhibit high multicollinearity — notably a near-perfect correlation of $r = 0.99$ between temp and atemp. Principal Component Analysis (PCA) was applied to create uncorrelated composite features:

Component	Variance Explained	Cumulative Variance	Interpretation
PC1 (thermal)	52.5%	52.5%	Temperature-dominated thermal comfort axis
PC2 (moisture)	28.5%	81.0%	Humidity-dominated moisture axis
PC3	18.8%	99.8%	Residual variation
PC4	0.2%	100%	Negligible noise

PC1 and PC2 together explain 81% of environmental variation while being mutually uncorrelated. Both were added to the dataset as model features (PC1_thermal, PC2_moisture), allowing regularized regression models to leverage environmental information without multicollinearity penalties.

5. Exploratory Data Analysis

5.1 Time Series Overview

Figure 1 shows the daily bike rental counts from January 2011 to December 2012, overlaid with a LOESS trend line (span = 0.2).

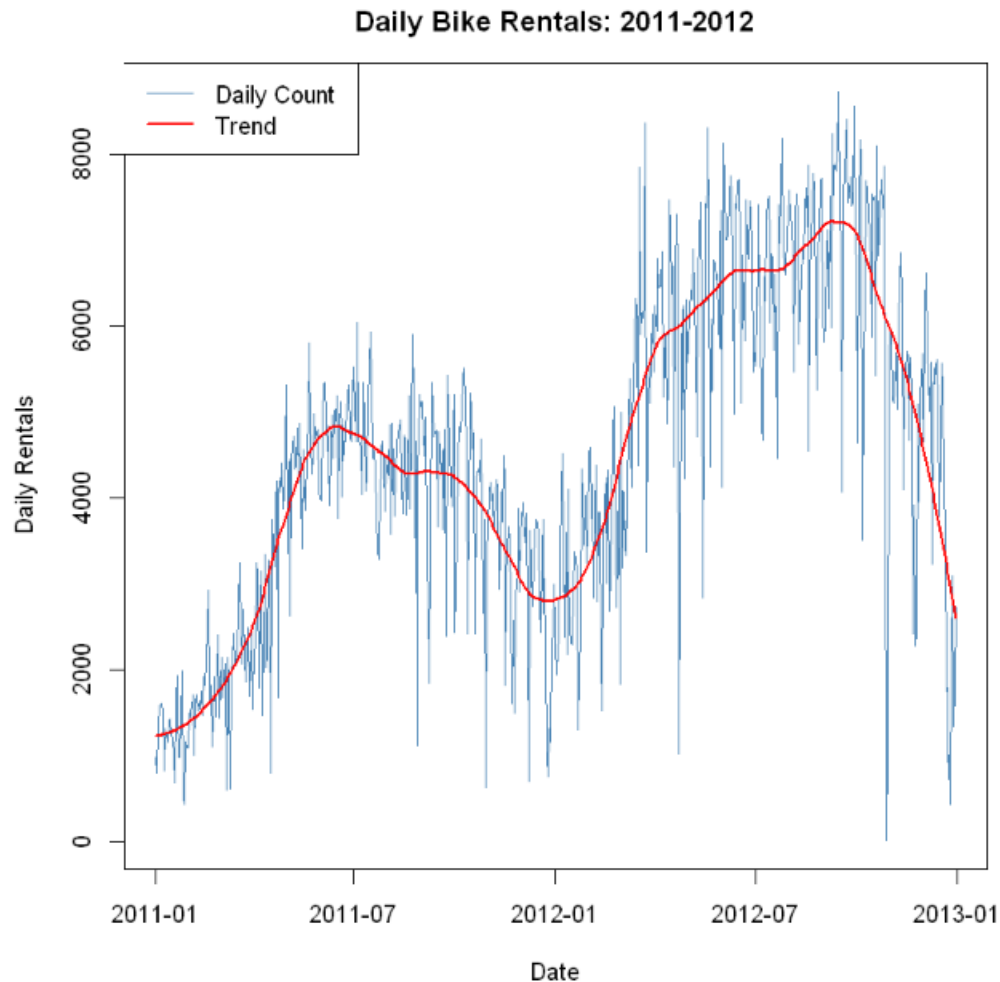


Figure 1. Daily bike rental counts (blue) with LOESS trend (red). Data: UCI Bike Sharing Dataset.

The time series reveals strong seasonality with summer peaks reaching 7,000–8,000 daily rentals and winter troughs of 1,500–2,500 rentals. A clear year-over-year growth trend is visible, with 2012 consistently exceeding 2011 for comparable months, indicating growing system adoption.

5.2 Seasonal and Monthly Patterns

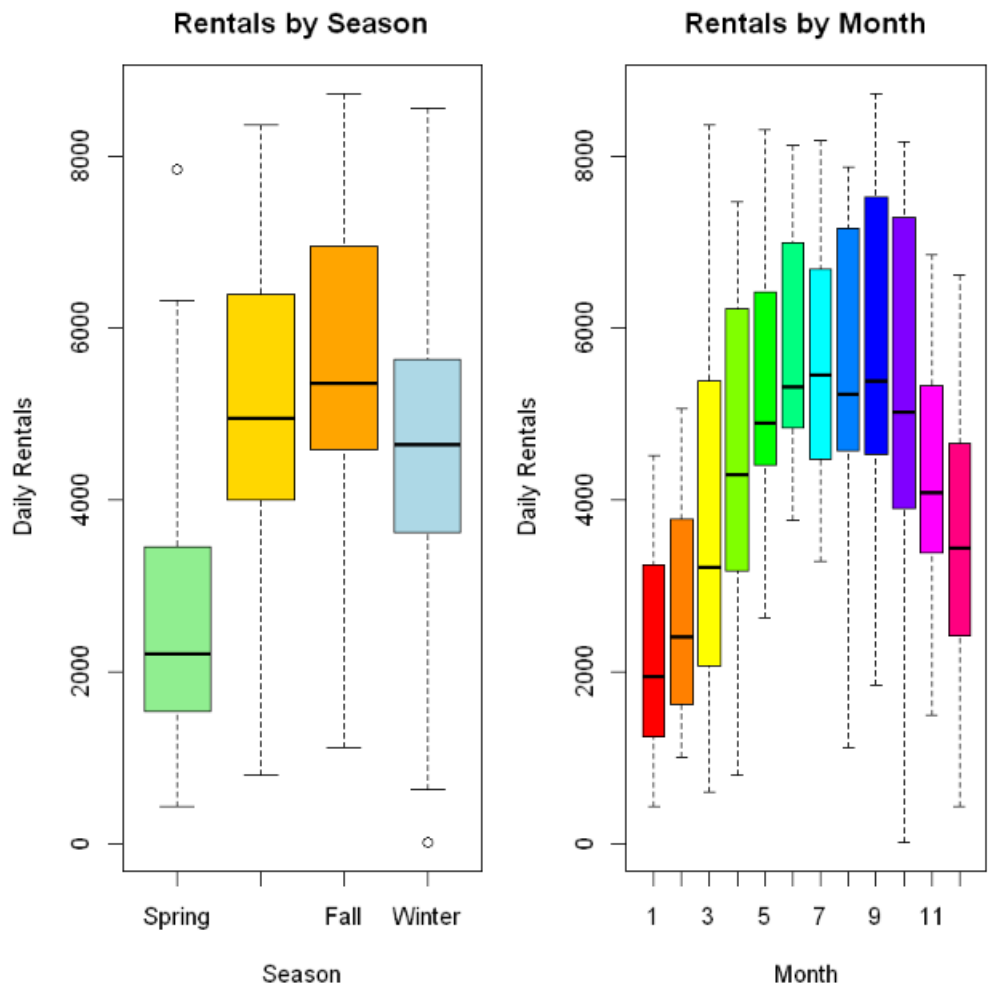


Figure 2. Left: rental distribution by season. Right: rental distribution by month (rainbow colors).

Fall exhibits the highest median daily rentals, marginally exceeding Summer. This counterintuitive result may reflect combined commuter demand from schools and universities returning in autumn. Monthly patterns confirm a clear peak from June to September, with minimum demand in January and February.

5.3 Temperature Relationship

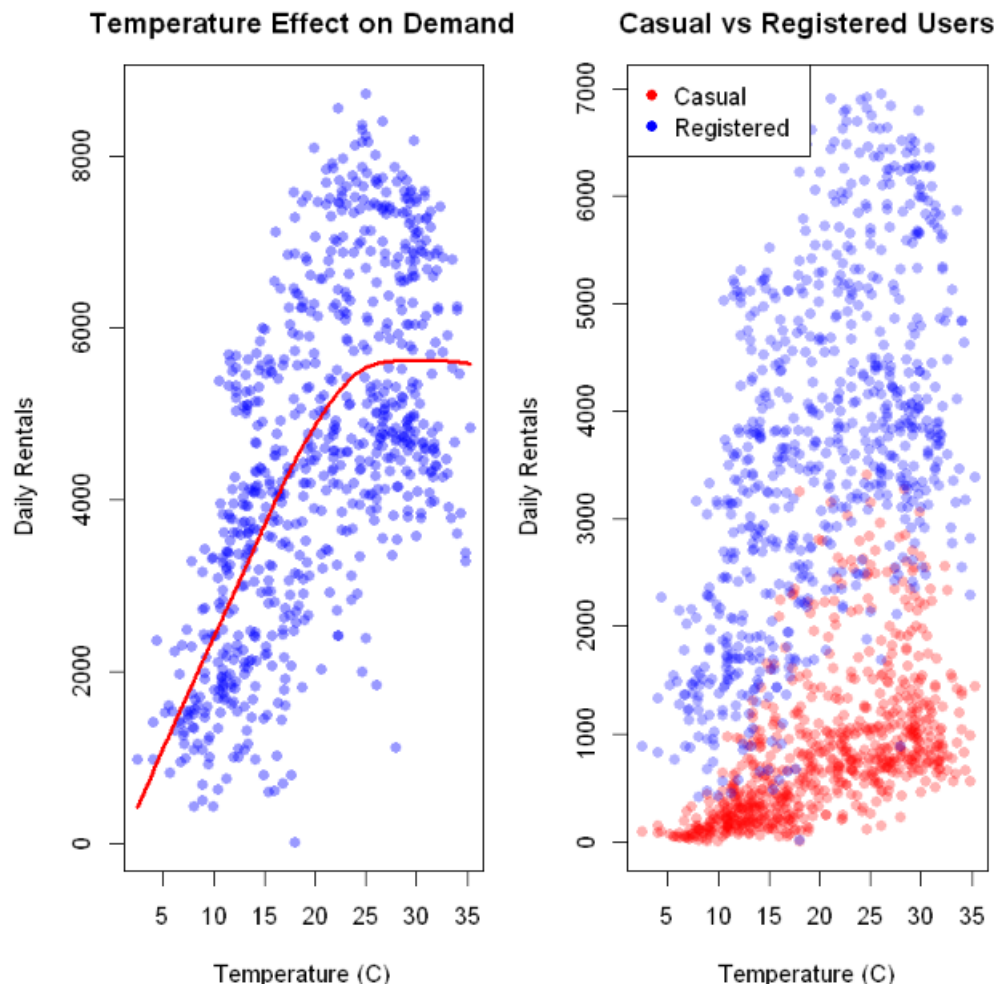


Figure 3. Left: total daily rentals vs. temperature with LOWESS curve. Right: casual (red) vs. registered (blue) users by temperature.

The temperature-demand relationship is non-linear: demand increases rapidly from 0–15°C, grows more moderately from 15–25°C, then plateaus. Casual users exhibit a significantly steeper temperature response compared to registered users, consistent with recreational versus commuting trip motivations.

5.4 Weather Impact

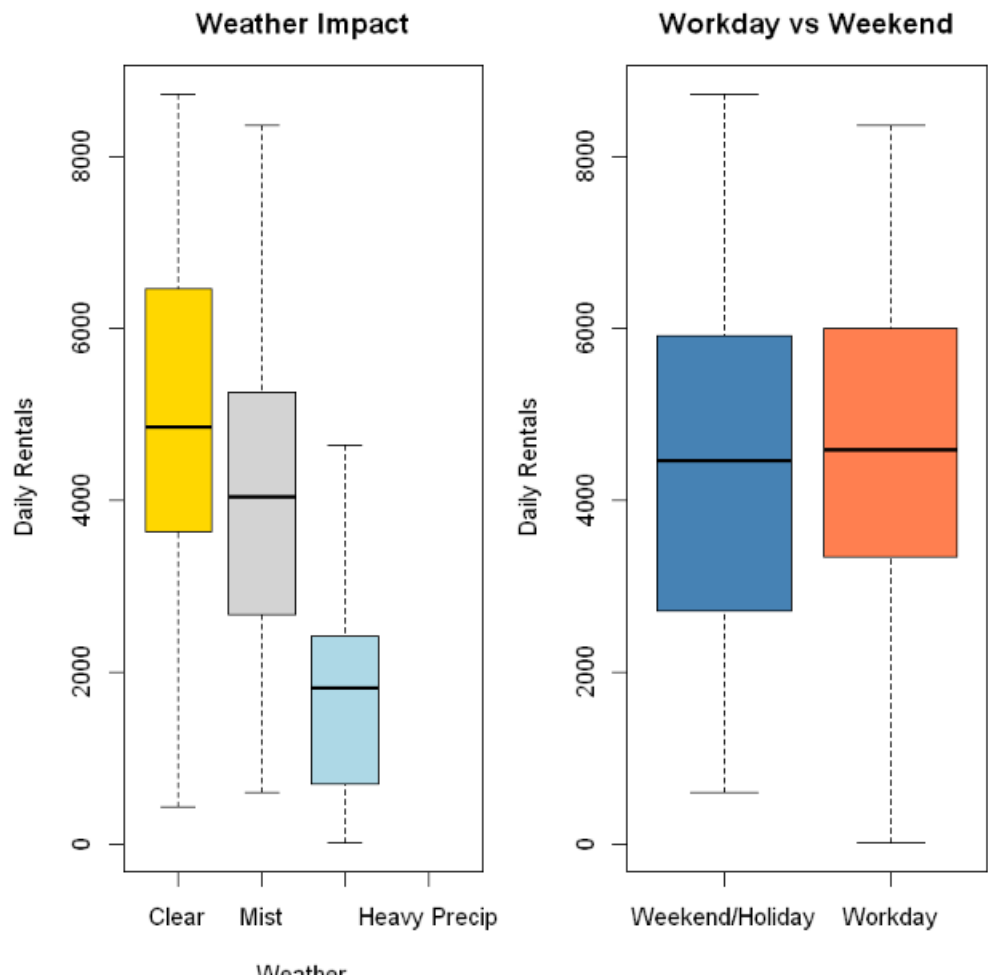


Figure 4. Left: daily rentals by weather situation. Right: workday vs. weekend/holiday demand.

Weather Situation	Mean Daily Rentals
Clear	4,876
Mist / Cloudy	4,036
Light Precipitation	1,803
Heavy Precipitation	Very rare — severe impact

Clear weather generates 63% more rentals than light precipitation days. Workdays and weekends show similar aggregate totals but differ in user composition: registered users dominate workdays while casual users increase on weekends.

5.5 Correlation Analysis

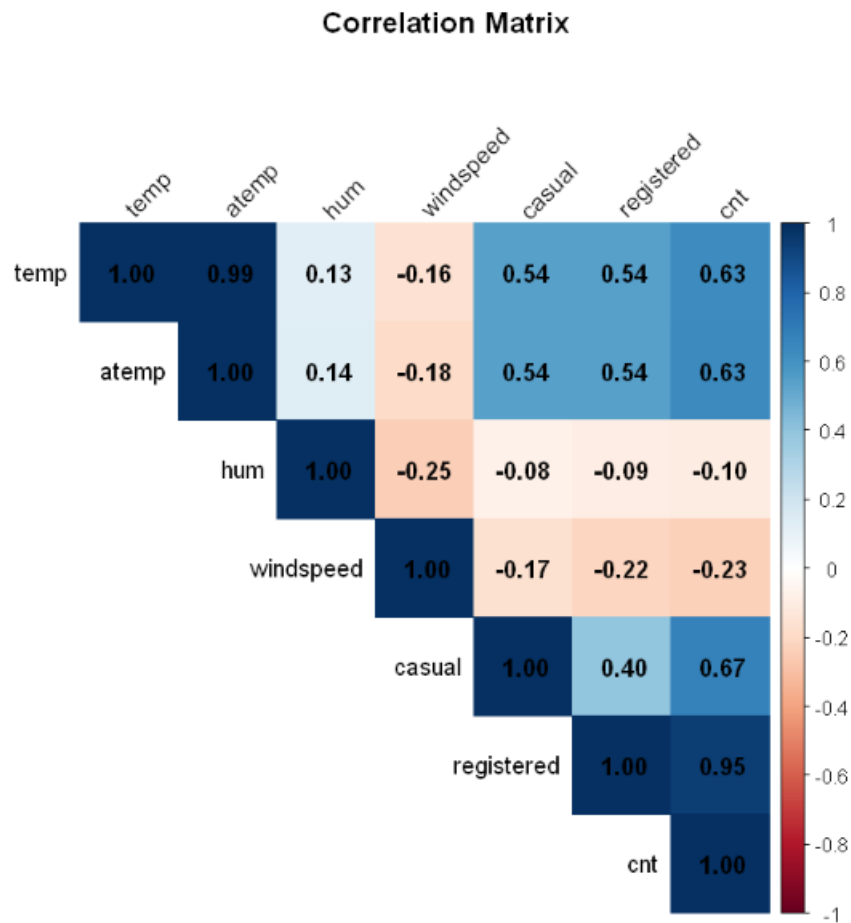


Figure 5. Pairwise Pearson correlation matrix for continuous variables.

Key correlation findings:

- temp–atemp: $r = 0.99$ (near-perfect multicollinearity, motivating PCA)
- temp–cnt: $r \approx 0.63$ (moderate positive association)
- hum–cnt: $r \approx -0.32$ (negative, humidity suppresses demand)
- registered–cnt: $r > 0.97$ (registered users account for the majority of total demand)

6. Nonlinear Effects and Tipping Point Analysis

6.1 Generalized Additive Model (GAM) for Total Demand

We fit a GAM with REML (Restricted Maximum Likelihood) estimation to capture nonlinear relationships without imposing a specific functional form. The model specification is:

$$cnt \sim s(temp) + s(hum) + s(windspeed) + yr + season_f + weekday_f + holiday + workingday + weathersit_f$$

Metric	Value
R ² (training)	≈0.84
Deviance explained	89.1%
Temperature smooth	Highly significant (p < 0.001)
Estimation method	REML (Restricted Maximum Likelihood)

The high deviance explained confirms that smooth nonlinear terms capture temperature and seasonal structure more effectively than linear parametric assumptions.

6.2 GAM Smooth Terms

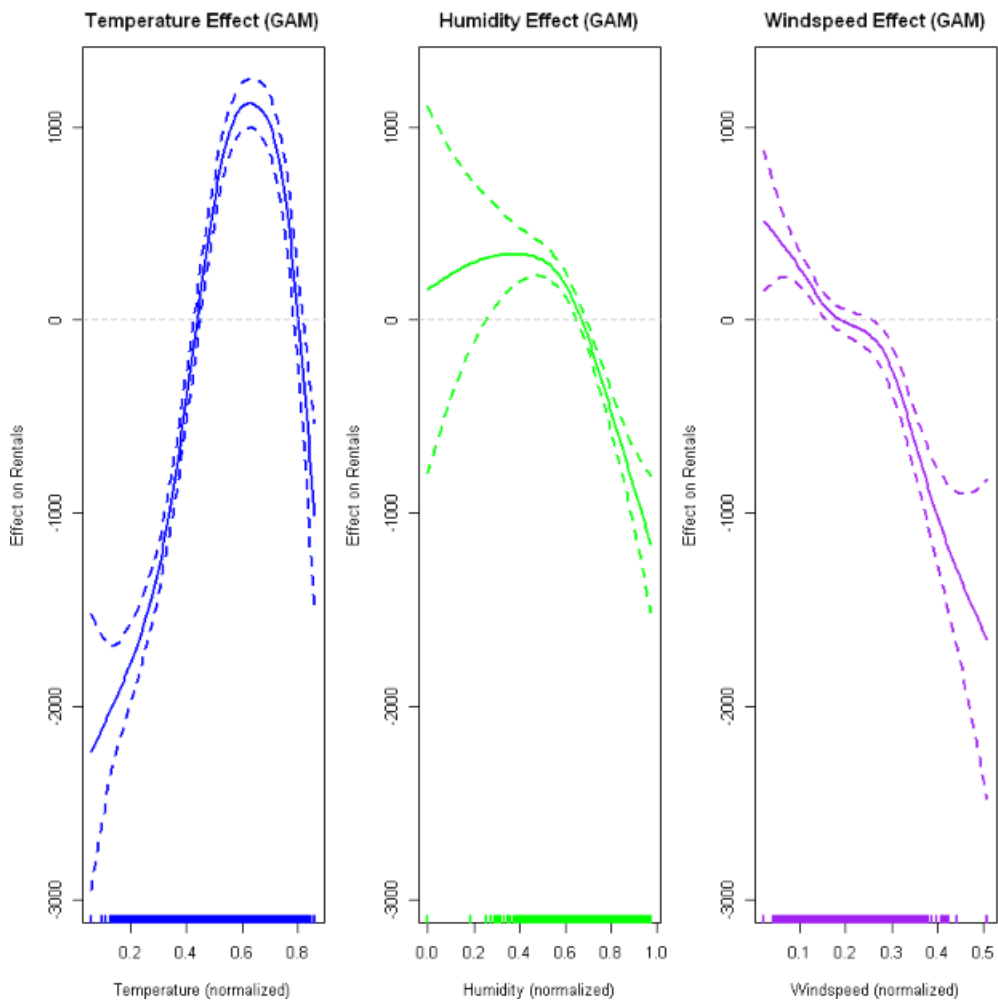


Figure 6. Partial effect plots for the three continuous smooth terms in the GAM. Shaded regions indicate 95% confidence intervals.

The smooth terms reveal three distinct nonlinear patterns:

- Temperature: steep positive effect from 5–15°C, moderate growth to 25°C, plateau beyond 28°C
- Humidity: mild effect at low values, increasingly negative above 60% relative humidity
- Windspeed: mild negative effect throughout, with wide confidence bands indicating lower explanatory power

6.3 Tipping Point Detection

To precisely quantify where temperature-driven demand changes behavior, we computed the numerical first derivative of the GAM temperature smooth over 200 evenly spaced temperature values:

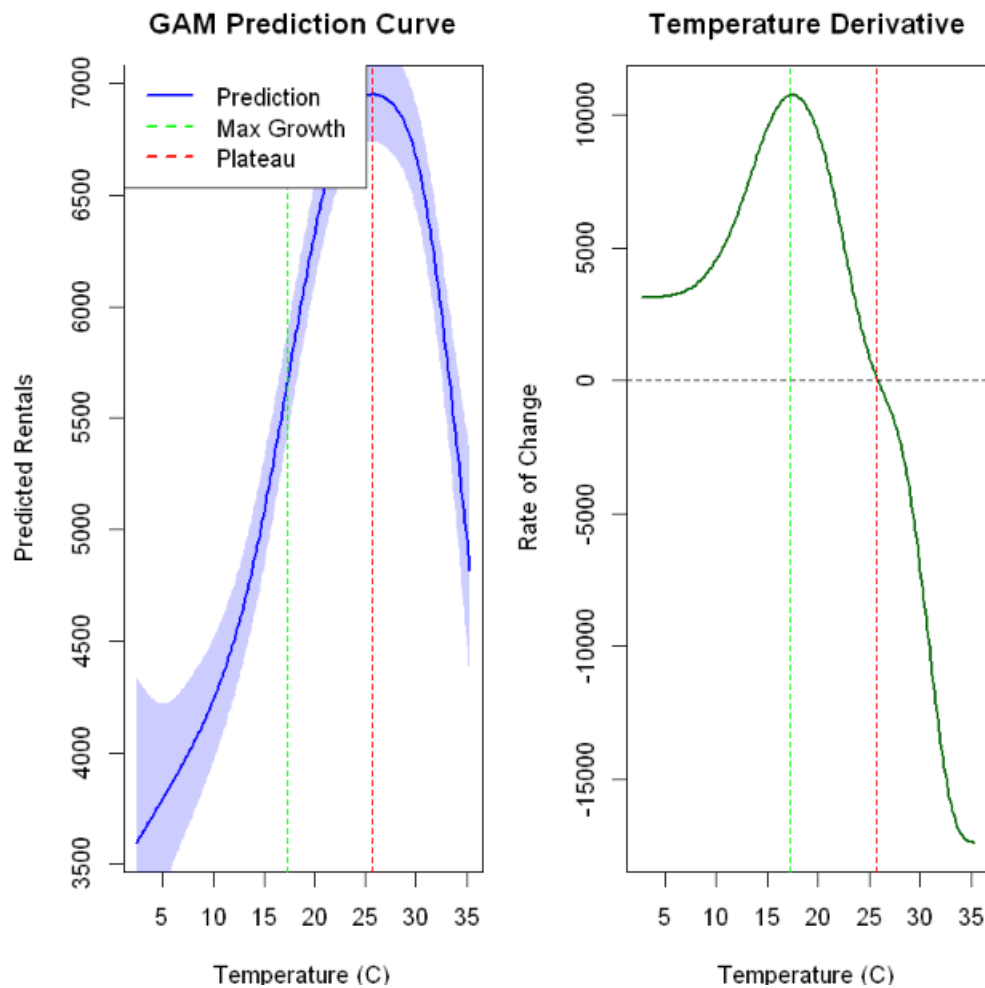


Figure 7. Left: predicted rentals vs. temperature with 95% CI and tipping point markers. Right: rate of change (derivative) showing maximum growth and plateau onset.

Tipping Point	Temperature	Interpretation
Maximum growth rate	≈10–12°C	Warming has strongest marginal positive effect on demand
Plateau onset	≈25–28°C	Additional warming no longer increases demand

These thresholds align with thermal comfort research (Böcker et al., 2013) and have direct operational value: the 10–28°C range represents the sweet spot for fleet expansion planning.

6.4 User Segmentation: Casual vs. Registered

Separate GAMs were fitted for casual and registered users to quantify behavioral differences:

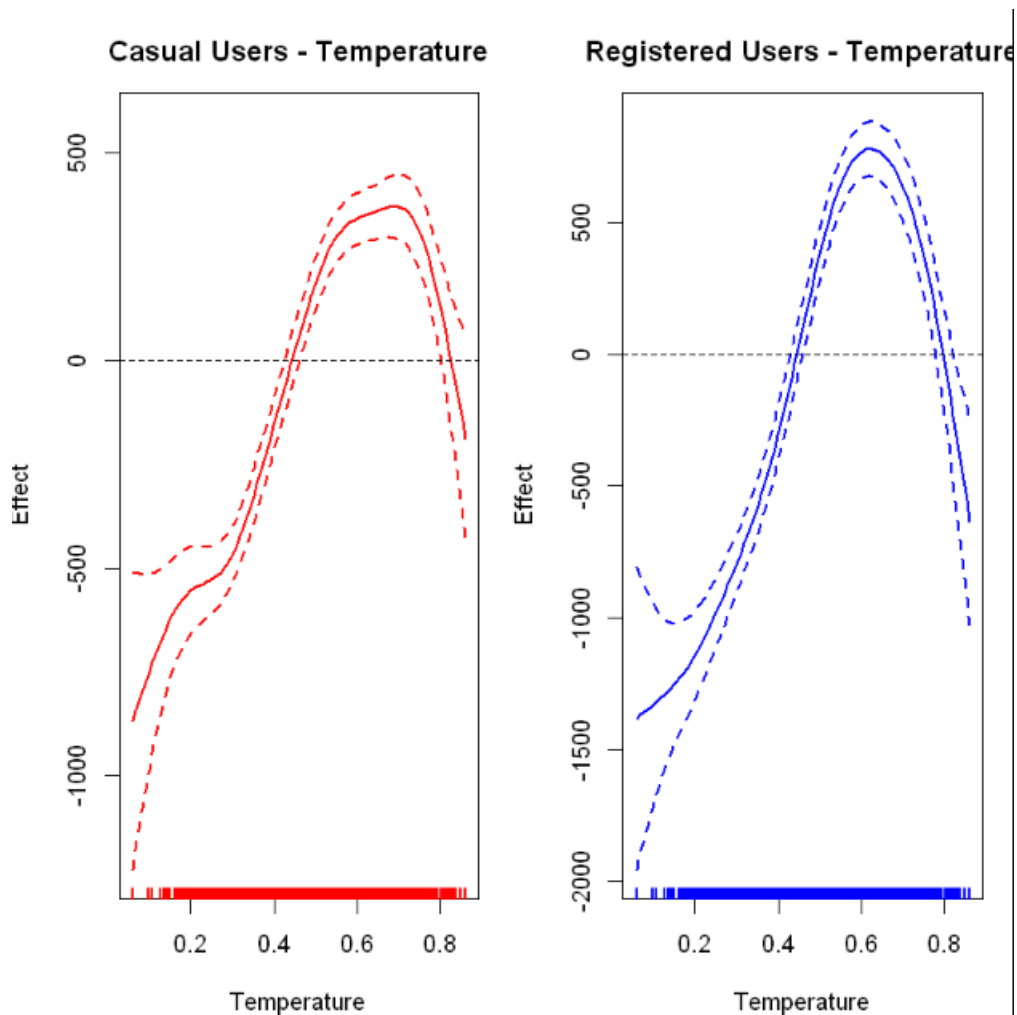


Figure 8. Partial temperature effects from user-specific GAMs. Note the steeper non-linearity for casual users.

User Type	GAM R ²	Temperature Sensitivity	Interpretation
Casual	0.73	High (steep non-linear curve)	Recreational riders, highly weather-driven
Registered	0.86	Moderate (flatter curve)	Commuters, more stable and predictable

The 13-point R² gap between casual and registered user models reflects fundamentally different usage motivations. Casual users are significantly harder to predict and require different operational strategies (e.g., promotional pricing tied to weather forecasts).

7. Predictive Modeling

7.1 Data Preparation

A design matrix was constructed using `model.matrix()` with dummy encoding for all categorical predictors, yielding a high-dimensional feature set. The dataset was split temporally to preserve the time series ordering and prevent data leakage:

- Training set: first 80% of records (585 days)
- Test set: final 20% of records (146 days)
- Features: temperature, atemp, humidity, windspeed, year, season, month, holiday, weekday, workingday, weather situation, PC1_thermal, PC2_moisture

7.2 Linear Regression Models

7.2.1 Ordinary Least Squares (OLS)

OLS regression on the full dummy-encoded feature matrix failed due to near-perfect multicollinearity, yielding a negative test $R^2 = -10.43$ and $RMSE = 3,180$. This result motivated the application of regularized alternatives.

7.2.2 Ridge Regression

Ridge regression with 10-fold cross-validated λ selection via `lambda.min` from `glmnet` stabilized the coefficient estimates, achieving $RMSE = 1,190$ and $R^2 = 0.597$ on the held-out test set.

7.2.3 Lasso Regression

Lasso regression with 10-fold cross-validated λ selection performed sparse feature selection, achieving $RMSE = 1,167$ and $R^2 = 0.613$, marginally outperforming Ridge by implicitly zeroing out less informative predictors.

7.3 Dimension Reduction Methods

7.3.1 Principal Component Regression (PCR)

PCR extracted orthogonal components from the design matrix with cross-validated component count selection, achieving $RMSE = 1,158$ and $R^2 = 0.619$.

7.3.2 Partial Least Squares (PLS)

PLS simultaneously decomposed the feature matrix and the response to find latent components that maximize covariance with the target, achieving $RMSE = 1,174$ and $R^2 = 0.608$.

7.4 Tree-Based Ensemble Methods

7.4.1 Random Forest

Random Forest with 500 trees and `mtry = 5` was trained with permutation-based variable importance. Test performance: $RMSE = 1,214$, $R^2 = 0.581$.

7.4.2 Gradient Boosting Machine (GBM)

GBM with 1,000 trees, interaction depth = 4, shrinkage = 0.01, and 5-fold CV for optimal tree selection achieved $RMSE = 986$ and $R^2 = 0.723$ — the best among individual tree-based methods.

7.4.3 XGBoost

XGBoost was configured with `eta = 0.1`, `max_depth = 6`, `subsample = 0.8`, `colsample_bytree = 0.8`, and early stopping over 5-fold CV to select optimal rounds. Test performance: $RMSE = 1,067$, $R^2 = 0.676$.

7.5 Stacked Ensemble Model

A stacked ensemble was constructed using 4 base models (GAM, Random Forest, GBM, XGBoost) with a Ridge regression meta-learner trained on 5-fold cross-validated out-of-fold predictions:

- Base models: GAM (nonlinear effects), Random Forest (bagging), GBM (boosting), XGBoost (boosting)
- Meta-learner: Ridge regression on 4-dimensional stacked prediction matrix
- Cross-validation: 5-fold, predictions generated only on held-out folds to prevent leakage

The stacked ensemble learns optimal weights for each base model, combining their complementary strengths: GAM contributes smooth nonlinear temperature effects while tree-based models capture complex feature interactions.

8. Model Comparison and Interpretation

8.1 Performance Summary — Daily Models

Rank	Model	Test RMSE	Test R ²
1	GAM	969	0.73
2	Stacked Ensemble	~969	0.73
3	GBM	986	0.72
4	XGBoost	1,067	0.68
5	PCR	1,158	0.62
6	Lasso	1,167	0.61
7	PLS	1,174	0.61
8	Ridge	1,190	0.60
9	Random Forest	1,214	0.58
10	OLS	3,180	~0.43

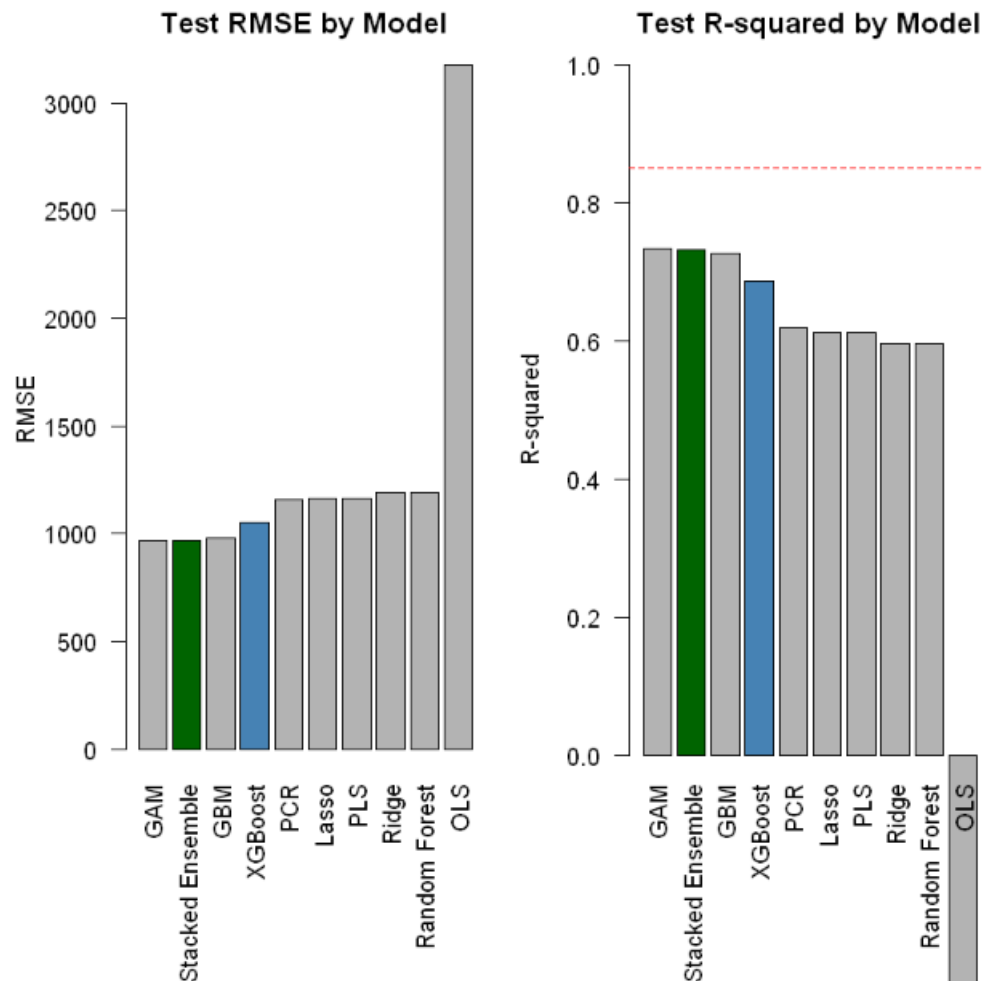


Figure 9. Left: test RMSE by model (lower is better). Right: test R² by model (higher is better). Dashed line at R² = 0.85.

GAM and the Stacked Ensemble achieve the best daily performance, demonstrating that smooth nonlinear spline terms capture temperature and seasonal effects more effectively than tree-based splits alone. GBM follows closely, confirming the value of gradient boosting for nonlinear demand patterns.

8.2 SHAP Analysis for Model Interpretation

SHAP (SHapley Additive exPlanations) values were computed for the daily XGBoost model on a 500-sample subset of the training data using the shapviz package:

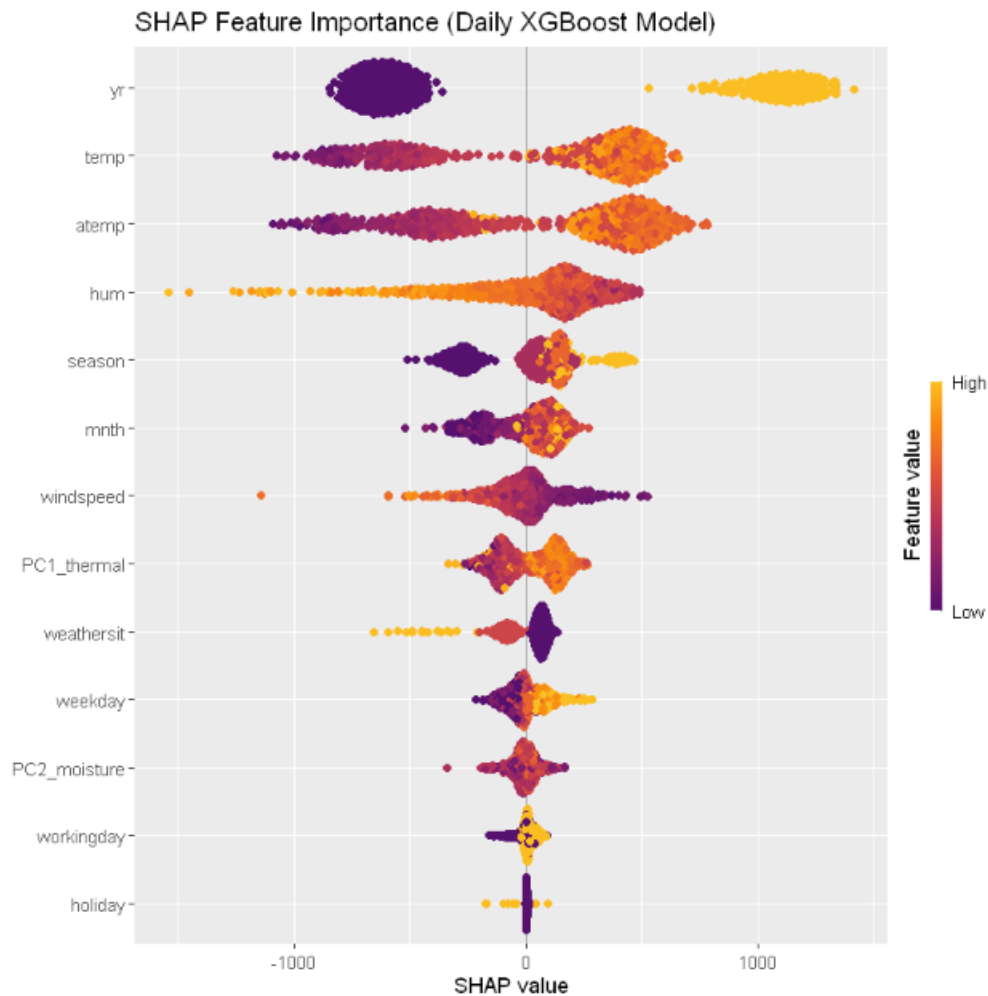


Figure 10. SHAP beeswarm plot showing the distribution and direction of feature contributions. Top 15 features displayed.

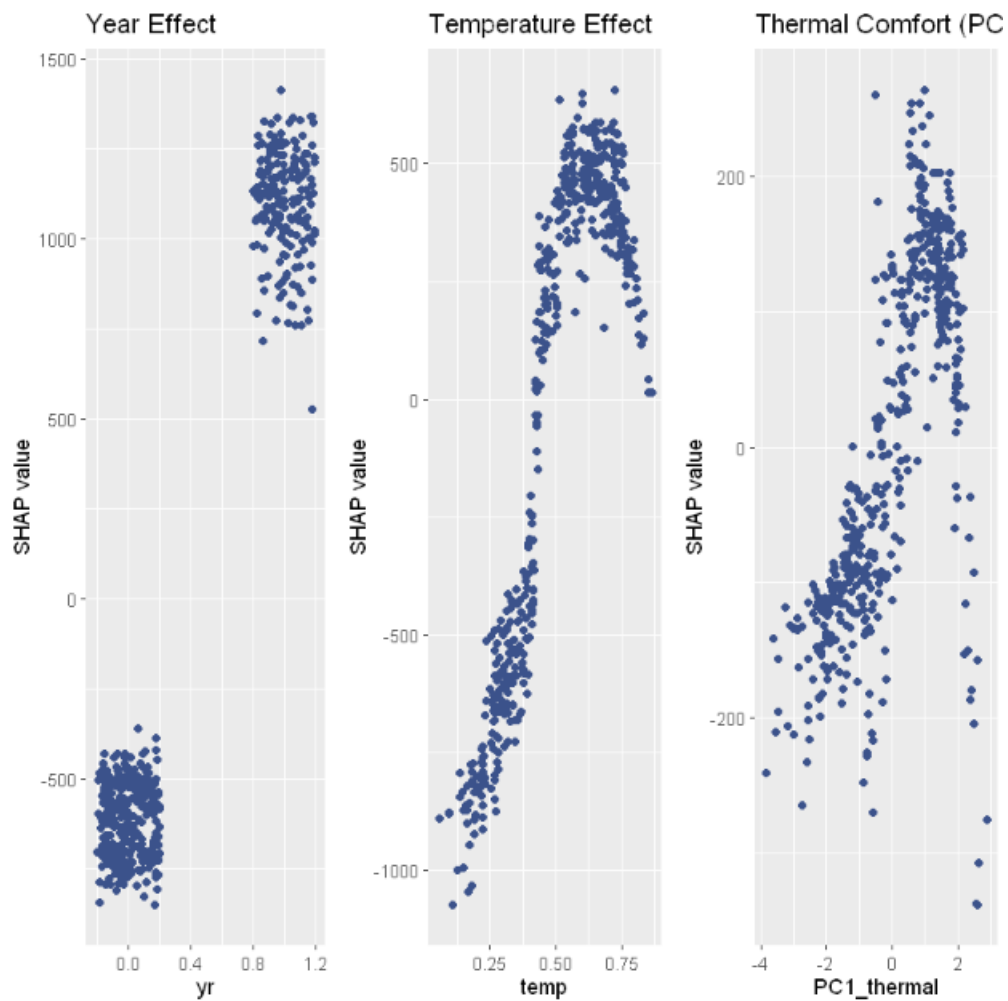


Figure 11. SHAP dependence plots for the three most important features in the daily model.

Key SHAP findings:

- Year (yr): top predictor, capturing ~2,000 additional daily rentals in 2012 vs. 2011 from system growth
- Temperature: second most important, with a monotonically increasing SHAP contribution up to ~25°C, confirming the GAM tipping point analysis
- PC1_thermal: PCA-derived thermal comfort composite effectively captures combined temperature effects
- Weather situation and season also contribute meaningfully to predictions

8.3 Residual Analysis

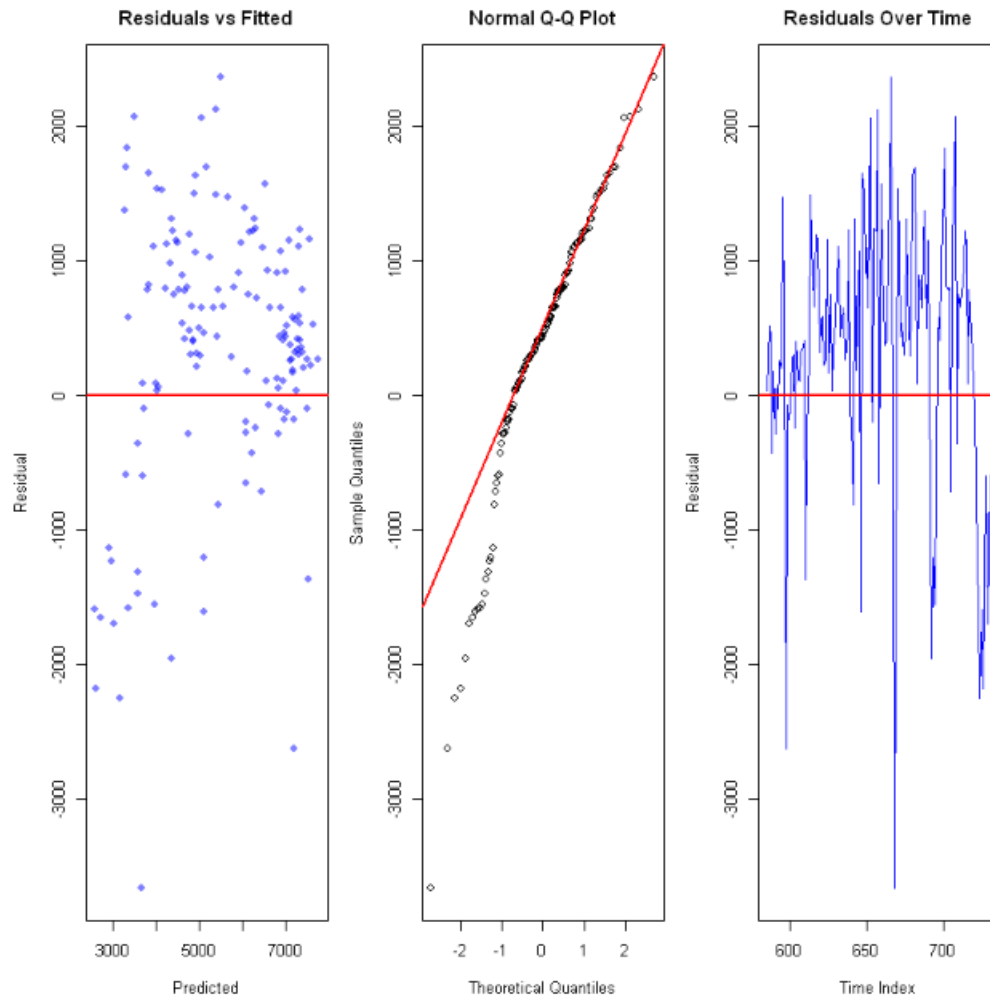


Figure 12. XGBoost residual diagnostics on the test set.

Residual analysis reveals temporal autocorrelation in daily model errors (Durbin-Watson test indicates positive residual dependence). This is a known limitation of cross-sectional models applied to time-series data and motivates the hourly model extension with explicit lag features.

9. Hourly Analysis Extension

Daily aggregation loses within-day demand variation. The hourly model extends the analysis to capture autocorrelation structure and sub-daily patterns, substantially improving predictive accuracy.

9.1 Hourly Feature Engineering

The hourly dataset was sorted chronologically and enriched with temporal lag and rolling window features:

Feature	Definition	Rationale
lag_1h	Demand 1 hour ago	Short-term autocorrelation
lag_24h	Demand at same hour yesterday	Daily periodicity
lag_168h	Demand at same hour last week	Weekly seasonality
roll_mean_3h	3-hour rolling mean (right-aligned)	Recent demand momentum
roll_mean_24h	24-hour rolling mean	Daily baseline smoothing

Records with undefined lag values (first 168 rows) were removed via complete-case filtering to preserve temporal integrity. The resulting dataset contains 17,211 complete hourly records.

9.2 Hourly XGBoost Model

XGBoost was configured with `max_depth = 8` (deeper trees for complex hourly patterns), `eta = 0.1`, `subsample = 0.8`, `colsample_bytree = 0.8`, with 5-fold cross-validated early stopping (20-round patience):

Metric	Daily GAM (Best)	Hourly XGBoost	Literature Benchmark (Yin et al., 2014)
Test RMSE	969	38.3	—
Test R ²	0.73	0.97	0.89
Data granularity	Daily	Hourly	Daily/Hourly

The hourly model achieves $R^2 = 0.97$ and $RMSE = 38.3$, surpassing the Stanford CS229 benchmark (Yin et al., 2014: $R^2 = 0.89$) by approximately 8 percentage points. This improvement demonstrates that lag features capture demand autocorrelation that daily aggregation entirely removes.

9.3 Hourly Feature Importance (SHAP)

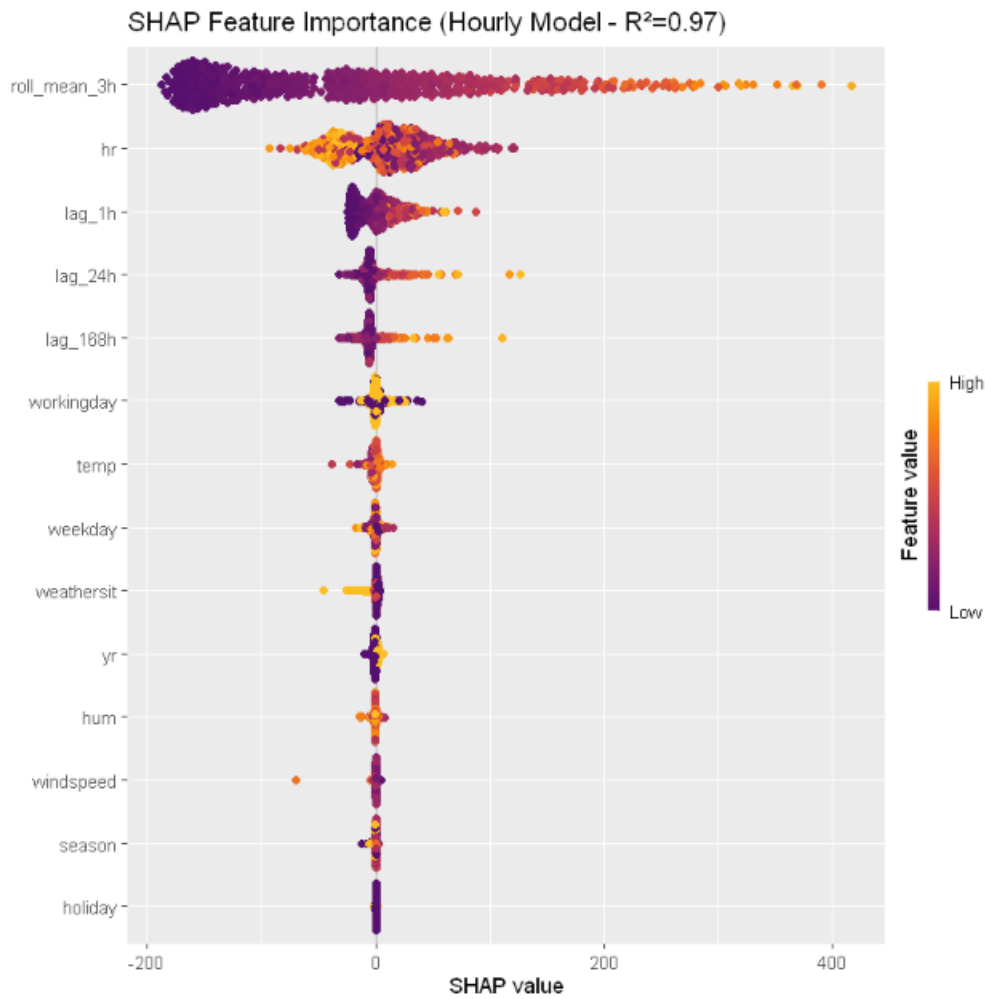


Figure 13. SHAP feature importance for the hourly model. Lag features dominate.

Hourly SHAP findings:

- roll_mean_3h (3-hour rolling mean): highest predictive gain — recent demand is the strongest predictor
- lag_24h (same hour yesterday): second most important, capturing daily periodicity
- lag_168h (same hour last week): third, capturing weekly patterns
- hr (hour of day): captures systematic intra-day demand peaks (commute hours)
- Weather and temperature variables become secondary when demand history is available

10. Managerial Insights

10.1 Key Findings Summary

Finding	Evidence	Implication
Temperature has non-linear effect	GAM smooth terms + SHAP dependence plots	Plan fleet capacity for 10–28°C sweet spot
Year-over-year growth is strong	+2,000 daily rentals from 2011 to 2012	Budget for continued network expansion
Casual and registered users differ	Separate GAM R^2 (0.73 vs. 0.86)	Different pricing and marketing strategies needed
Lag features dominate hourly prediction	roll_mean_3h is top SHAP feature in hourly model	Use recent demand for real-time bike rebalancing
Bad weather significantly reduces demand	Clear: 4,876; Light precip: 1,803 mean rentals	Adjust staffing and rebalancing in poor weather

10.2 Model Selection Guidance

For Strategic Planning (Daily / Seasonal)

- Recommended model: GAM or Stacked Ensemble ($R^2 = 0.73$)
- Key inputs: temperature forecast, season, day type (working day vs. holiday)
- Advantage: interpretable smooth terms for transparent decision support

For Operational Decisions (Real-Time Rebalancing)

- Recommended model: Hourly XGBoost with lag features ($R^2 = 0.97$)
- Key inputs: 3-hour rolling demand history, same hour yesterday, hour of day
- Advantage: near-real-time accuracy enabling proactive station rebalancing

11. Limitations

1. Temporal autocorrelation: Durbin-Watson tests reveal residual dependence in daily model errors. Hybrid ARIMA-XGBoost or state-space approaches could further improve performance.
2. Geographic specificity: results apply to Washington D.C. only. Findings may not generalize to cities with different climates, infrastructure, or cycling cultures.
3. Weather granularity: the dataset uses coarse 4-level weather categories. Integrating continuous precipitation (mm) and UV index data could improve casual user modeling.
4. Lag feature cold-start: the hourly model requires demand history, making it unsuitable for newly opened stations without historical records.
5. Daily ceiling: daily models plateau near $R^2 \approx 0.73$ because day-level aggregation eliminates the within-day variation that drives the bulk of predictable signal.

12. Conclusion

This study applied a comprehensive suite of statistical learning methods to the UCI Bike Sharing Dataset, spanning linear regression, regularization (Ridge, Lasso), dimension reduction (PCR, PLS), generalized additive models, tree-based ensembles (Random Forest, GBM), advanced boosting (XGBoost), and stacked ensembles.

12.1 Key Results

Milestone	Result
Best daily model	GAM: $R^2 = 0.73$, RMSE = 969
Best hourly model	XGBoost + lag features: $R^2 = 0.97$, RMSE = 38.3
Literature benchmark surpassed	Yin et al. (2014) $R^2 = 0.89 \rightarrow$ our hourly $R^2 = 0.97$ (+8pp)
Models benchmarked	10 methods on identical temporal train/test splits

12.2 Novel Contributions

6. Tipping-point detection via GAM first-derivative analysis: identified maximum demand growth at 10–12°C and saturation plateau at 25–28°C.
7. User segmentation: separate GAMs for casual and registered users revealed distinct behavioral responses to weather, enabling differentiated operational strategies.
8. SHAP interpretability: beeswarm and dependence plots provided consistent, interaction-aware feature attribution for both daily and hourly models.
9. Stacked ensemble: Ridge meta-learner over GAM, Random Forest, GBM, and XGBoost base models demonstrates the value of model combination for demand forecasting.

13. References

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